Momo Kihara

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Education

University of California, San Diego — *La Jolla, CA*
B.S. and B.A. — *Expected June 2029*

- First Major: Cognitive Science with a specialization in Design and Interaction
- Second Major: Speculative Design
- Minor: Computational Social Science
- GPA 3.93/4.00; Provost Honors (Fall 2025, Winter 2026, Spring 2026)

Marist Brothers International School — *Hyogo, Japan*
International Baccalaureate Bilingual Diploma (English and Japanese) — *June 2025*
WASC (Western Association of Schools and Colleges)-accredited High School Diploma

Interests

My research investigates how experience, cultural knowledge, and learned structure shape internal perceptual and cognitive representations, and how these representations can be computationally modeled to predict perception and behavior. My previous work examines possible acquired synesthesia in competitive karuta players, using the distribution of reported synesthetic colors in CIE L*a*b* space to characterize how phonological, orthographic, and semantic features shape color associations.

Building on this work, I am developing cross-corpus predictive models of synesthetic color associations, trained on approximately 1,800 color responses to *Hyakunin Isshu* stimuli and validated on newly collected *Man'yōshū* responses, using hierarchical (multilevel) models that predict full color-response distributions. I am also interested in extending this framework from perception to action: modeling how biological constraints and authored goals combine to produce believable decision-making in animals, animated characters, and interactive robotic characters.

My research interests include:

- Computational Cognitive Science & Perceptual Modeling — Modeling how experience and learning shape internal representations and predict perceptual and behavioral outcomes.
- Synesthesia, Perceptual Learning & Cultural Expertise — Investigating how cultural expertise and phonological, orthographic, and semantic features shape acquired synesthetic color associations, using distributional analysis in CIE L*a*b* color space.
- Active Inference & Models of Decision-Making — Modeling sequential decision-making with active inference (POMDP generative models, expected free energy), and estimating individual preference and precision parameters from animal choice behavior.
- Computational Character Cognition — Investigating how animated and robotic characters combine biological behavioral dynamics with authored personality, by testing whether models that blend data-driven animal dynamics with descriptor-based character preferences better predict held-out, annotated behavioral sequences.

Research Experience

Kwansei Gakuin Institute of Kansei Value Creation — *Hyogo, Japan*
Student Researcher, Prof. Noriko Nagata Lab. Advisor: Dr. Daisuke Hamada — *April 2021 – Present*

- Began research in high school; conducting a fully funded, independent research project supported by the Japan Science and Technology Agency.
- Utilize MATLAB for screening tests and R for statistical analysis of 1,800 synesthetic color samples, spatial statistics, and first-authoring academic papers.

RIKEN Center for Brain Science, Statistical Mathematics Unit — *Saitama, Japan*
Student Researcher. Advisor: Dr. Takeru Matsuda — *July 2026 – Expected March 2028*

- Collaborating to study a cross-corpus prediction framework for synesthetic color associations using ~1,800 color responses to Hyakunin Isshu stimuli; modeling grapheme- and sound-color CIE L*a*b* distributions from phonological, orthographic, and semantic features.

- Planning to utilize multivariate hierarchical (multilevel) models that predict full color-response distributions, with poem-level cross-validation and prospective validation on newly collected Man'yōshū responses.

Publications

Kihara, M., Hamada, D., and Nagata, N. (2024, March 9). *Possible acquired synesthesia underlying the consistency of favorable colors: A case study of grapheme–color synesthesia and sound–color synesthesia among competitive karuta players.* In Proceedings of the 10th International Symposium on Affective Science and Engineering (Online). https://doi.org/10.5057/isase.2024-C000046

(In Prep.)

Kihara, M., Hamada, D., and Nagata, N. (in preparation). *A spatial statistical mapping of synesthetic color patterns in competitive karuta: Implications for acquired synesthesia.* Manuscript in preparation for submission to *Consciousness and Cognition* (Elsevier).

Presentations

Kihara, M. "Neural and Representational Signatures of Synesthetic Color Associations in Competitive Karuta Players" (poster). *RIKEN CBS Summer Program, Saitama, Japan, August 2026.*

Kihara, M. "When Culture Colors the Mind: Possible Acquired Synesthesia and Consistent Color Associations among Competitive Karuta Players." *Mentored Undergraduate Research and Applied Learning Symposium (MURALS), La Jolla, CA, May 28, 2026.*

Kihara, M. "What Color Is Amatsukaze? — Color Associations Evoked by Karuta Readings and Cards and Their Potential for Acquisition." *Annual General Assembly of the All Japan Karuta Association, Tokyo, Japan, June 7, 2025.*

Kihara, M. "The Potential for Acquired Synesthesia in Competitive Karuta: A Spatial Statistical Analysis of Synesthetic Colors associated with Individuals, Sound, and Nouns" (poster and oral). *Japan Science & Engineering Challenge 2024 (Japan preliminaries for ISEF 2025), hosted by The Asahi Shimbun, December 2024.*

Kihara, M. "The Potential for Acquired Synesthesia in Competitive Karuta: A Spatial Statistical Analysis of Synesthetic Colors" (poster and oral). *National Science Conference 2024, hosted by the Japan Science and Technology Agency, October 2024.*

Kihara, M., Hamada, D., and Nagata, N. "Spatial Analyses of Synesthetic Color Distribution Associated with a Japanese Card Game: A Comparative Study of Innate and Acquired Synesthetes" (poster). *UW Summer STEM Undergraduate Research Poster Session, Seattle, WA, August 14, 2024.*

Kihara, M., Hamada, D., and Nagata, N. "Possible Acquired Synesthesia Underlying the Consistency of Favorable Colors — A Case Study of Grapheme-color Synesthesia and Sound-color Synesthesia among Competitive Karuta Players" (oral). *International STEAM Symposium for Young Researchers, hosted by Kobe University, April 2024.*

Kihara, M., Hamada, D., and Nagata, N. "Possible Acquired Synesthesia Underlying the Consistency of Favorable Colors — A Case Study of Grapheme-color Synesthesia and Sound-color Synesthesia among Competitive Karuta Players" (oral). *International Symposium on Affective Science and Engineering 2024, Online, March 8, 2024.*

Kihara, M., Hamada, D., and Nagata, N. "The World of Synesthesia in Competitive Karuta" (poster). *Hyogo Regional Selection, 67th Japan Student Science Award, hosted by Yomiuri Shimbun, October 2023.*

Kihara, M. "Grapheme–color Synesthesia and Sound-color Synesthesia in Japanese Traditional Competitive Poetry Card Game" (poster). *8th Science Conference in Hyogo, hosted by Kobe University, July 2022.*

Research Grants

KAKENHI, Japan Society for the Promotion of Science *June 2025 – March 2028*
"Research for Competitive Karuta using Data Science." ¥6,500,000. Role: research team member.

Japan Science & Engineering Challenge 2024, Asahi Shimbun *December 2024*
"The Potential for Acquired Synesthesia in Competitive Karuta: Mapping Synesthetic Color Patterns with Spatial Statistical Analysis " ¥500,000.

Research-Oriented On-site Training Program, Japan Science and Technology Agency *April 2023 – March 2024*
"An Exploratory Study on the Acquisition of Synesthesia through Competitive Karuta." ¥300,000.

Fellowships & Awards

Yanai Tadashi Foundation Scholar *April 2025*
Yanai Tadashi Foundation. Full four-year undergraduate scholarship ($115,000/year) for academic merit and leadership potential.

Commendation for Science and Technology *December 2024*
Awarded by the Minister of Education, Culture, Sports, Science, and Technology (MEXT). 1st place with Overall Best Award and 1st place, Best Presentation Award, Japan Science & Engineering Challenge 2024 (Asahi Shimbun); offered to represent Japan at Regeneron ISEF 2025.

Commendation for Science and Technology *October 2024*
Awarded by the Minister of Education, Culture, Sports, Science, and Technology (MEXT). 1st place with Overall Best Award and 1st place, Best Presentation Award (Student Choice), National Science Conference 2024 (Japan Science and Technology Agency).

Hyogo Governor's Award *October 2023*
Hyogo Regional Selection, 67th Japan Student Science Award (Yomiuri Shimbun), Japan preliminaries for ISEF 2024.

Industry Experience

Recruit Holdings Co., Ltd. *Tokyo, Japan*
Product Management & Data Analytics Intern *July – August 2026*
- Led quantitative research and modeling (TAM–SAM–SOM, 3-year P&L with sensitivity analyses) on a 3-person team; designed a pricing model informed by 7 primary studies (n = 24–805) and interviews; analyzed 138K+ locations with SQL/BigQuery geospatial functions.

SoftBank Group Corp. *Tokyo, Japan*
Consumer Marketing & Digital Strategy Intern *August – September 2026*
- Applied persona definition and customer-journey mapping to B2C telecom campaigns; built a decision-support dashboard in Google Apps Script.

Additional Experience

Cognitive Science Student Association *La Jolla, CA*
Software Engineer & UI Designer *September 2025 – January 2026*
- Designed UI/UX for a mental health app; collaborated on photoplethysmography (PPG)-based algorithms for stress and attentional-state estimation.

The Catalyst at UCSD *La Jolla, CA*
Editorial Intern & Illustrator *January 2026 – Expected June 2027*
- Contribute to an academic publication and newsletter showcasing undergraduate research, in collaboration with the Undergraduate Research Hub.

Skills

Research: Spatial statistics, questionnaire design and analysis, geospatial analysis
Computer: R, MATLAB, Python, SQL, BigQuery, Git, Google Apps Script, Figma, Photoshop
Language: Japanese (native), English (IELTS 8.0), Korean (TOPIK 5), Spanish (elementary)

References

Dr. Noriko Nagata
Research mentor
Kwansei Gakuin University, Japan
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Dr. Masayuki Itoh
Program mentor
Kobe University, Japan
itoh@radix.h.kobe-u.ac.jp